Practical 18: SQLite with ListView or RecyclerView

Source: MODULE 2.pdf | Source pages: 65–71

Aim

To develop and test an Android application demonstrating sqlite with listview or recyclerview.

Steps to Perform the Practical

1. Create a project for displaying database records in a list.

2. Set up the SQLite database and CRUD methods using the source practical.

3. Add a ListView or RecyclerView to the activity layout, according to the supplied code.

4. Read records from SQLite and place them in an adapter.

5. Refresh the adapter after adding, changing or deleting a record.

6. Run the app and verify that database records appear in the list and update after changes.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

11. Develop an Android application integrating SQLite with ListView or RecyclerView.

Project Name : SQLiteCRUDDemo
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">

   <EditText
       android:id="@+id/etName"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:hint="Enter Name" />

   <Button
       android:id="@+id/btnInsert"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:text="Insert" />
```

```
    <Button
        android:id="@+id/btnView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="View Data" />
    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
</LinearLayout>
```

Create New Java Class in Package Folder

Name as: DBHelper

```
package com.example.sqlitecruddemo;


import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;


import java.util.ArrayList;


public class DBHelper extends SQLiteOpenHelper {


    public DBHelper(Context context) {
```

```
        super(context, "StudentDB", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE student(name TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        // Database upgrade code goes here
    }

    // Insert Data
    public void insertData(String name) {
        SQLiteDatabase db = this.getWritableDatabase();

        ContentValues values = new ContentValues();
        values.put("name", name);

        db.insert("student", null, values);
    }

    // Get All Data
    public ArrayList<String> getData() {
```

```
        ArrayList<String> list = new ArrayList<>();


        SQLiteDatabase db = this.getReadableDatabase();


        Cursor cursor = db.rawQuery("SELECT name FROM student", null);


        while (cursor.moveToNext()) {

            list.add(cursor.getString(0));

        }


        cursor.close();


        return list;

    }

}
```

MainActivity.java

```
package com.example.sqlitecruddemo;


import android.os.Bundle;

import android.widget.ArrayAdapter;

import android.widget.Button;

import android.widget.EditText;

import android.widget.ListView;

import android.widget.Toast;
```

```java
import androidx.appcompat.app.AppCompatActivity;


import java.util.ArrayList;


public class MainActivity extends AppCompatActivity {


   EditText etName;
   Button btnInsert, btnView;
   ListView listView;


   DBHelper dbHelper;


   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);


       setContentView(R.layout.activity_main);


       etName = findViewById(R.id.etName);
       btnInsert = findViewById(R.id.btnInsert);
       btnView = findViewById(R.id.btnView);
       listView = findViewById(R.id.listView);


       dbHelper = new DBHelper(this);
```

```
        // Insert Data
        btnInsert.setOnClickListener(v -> {

            String name = etName.getText().toString();

            dbHelper.insertData(name);

            Toast.makeText(
                    this,
                    "Data Inserted",
                    Toast.LENGTH_SHORT
            ).show();

            etName.setText("");
        });

        // View Data
        btnView.setOnClickListener(v -> {

            ArrayList<String> list = dbHelper.getData();

            ArrayAdapter<String> adapter = new ArrayAdapter<>(
                    this,
                    android.R.layout.simple_list_item_1,
                    list
            );
```

```
            listView.setAdapter(adapter);

        });

    }

}
```

